

Solution of Equations

1.2 GENERAL PROPERTIES

Example 1.1. Solve the equation $2x^3 + x^2 - 13x + 6 = 0$.

By inspection:-

$x = 2$ satisfies the given eqⁿ

∴ 2 is its root ✓

$(x-2)$ is a factor

Dividing the polynomial by $(x-2)$ ✓

$$\underline{2x^2 + 5x - 3 = 0}$$

Solving the quadratic eqⁿ

$$x = \frac{-5 \pm \sqrt{5^2 - 4 \cdot 2 \cdot (-3)}}{2 \cdot 2}$$

$$= \frac{-5 \pm \sqrt{25 + 24}}{4}$$

$$= \frac{-5 \pm 7}{4} = -3, \frac{1}{2}$$

Hence, the roots of the given eqⁿ are 2, -3, $\frac{1}{2}$

Synthetic division

$$\begin{array}{r|rrrr} 2 & 2 & 1 & -13 & 6 \\ & & +4 & 10 & -6 \\ \hline & 2 & 5 & -3 & 0 \end{array}$$

$$\underline{2x^2 + 5x - 3}$$